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9 January 1974
SUPERSEDING
MIL-C-46272(ORD)
30 May 1965

MILITARY SPECIFICATION

CARTRIDGE, DELAY, M46 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, Delay, M46 Assembly for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-B-550	- Barium Chromate
MIL-B-51092	- Boron, Amorphous, Powder
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloy
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-P-46994	- Pellets, Boron, Potassium Nitrate

STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Inspection Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O Rings
MS-28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Armament Command

D8594075	- Cartridge, Delay, M46 Assembly
D11730536	- Delay Element, Delay 2 Seconds, Assembly
C8845190	- Carton, Packing, Ammunition for Cartridge, Delay, M46
C8845191	- Box, Packing, Ammunition for Cartridge, Delay, M46
C9227731	- Alternative, Carton, Packing for Cartridge, Delay,M46
IEL-8594075	- Inspection Equipment List for Cartrid e, Delay, M46

(Copies of specifications, standards, drawings and publications required by suppliers in connnection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assem.lies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without ev dence of any d sintegration of the delay charge, primer, igniter or propellant or derangement of components.

3.4 Primer sensitivity.-The primer shall fire when drop tested at an energy level of 26 inch ounces in a sectioned cartridge case, void of dela charge, igniter and propellant charge.

3.5 Delay element assembly.-The delay element components shall be combined to produce an a sembly which conforms to the requirements of Drawing D11730536.

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3.6 First article sample.-Unless otherwise directed by the Contracting Officer, a first article sample shall be required.

3.7 Functional.

3.7.1 The delay element shall not fail to fire.

3.7.2 Delay charge.-The delay charge shall consist of pellets, Boron Potassium Nitrate Type II-A as specified in MIL-P-46994, Barium Chromate as specified in MIL-B-550 and Boron as specified in MIL-B-51092 in the proportion and quantity fixed by the contractor to produce the delay time as required in Table I (see 6.2, 6.3, and 6.4).

TABLE I

Temperature	Delay Time Seconds (see 6.3)
-65° to 70°F	2.4 max.
65° to 75°F	1.85 to 2.1
160° to 165°F	1.65 min.

3.7.3 The cartridge shall not fail to fire.

3.7.4 Ballistics.-The cartridge when tested in an M4 or M6A1 Initiator, shall comply with the requirements prescribed in Table II.

TABLE II

Temperature	Pressure (psi) 1/		Delay Time Seconds	
	Minimum	Maximum	Minimum	Maximum
-65° to -70°F	1300	2200		2.5
65° to 75°F	1300	2200	1.8	2.2
160° to 165°F	1300	2200	1.6	

1/ Pressure values shall be produced in a one-cubic inch pressure station at the end of 3 feet of hose (Size No. 4) in accordance with MS-28741. (See 6.5).

3.8 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

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3.8.1 Metal defects.-The metal parts shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.8.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the Quality Assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 Classification of inspections.-The inspection requirements specified herein are classified as follows:

(1) First Article Inspection
(2) Quality Conformance Inspection.

4.3 First article sample.-At the beginning of production, a sample representative of the production materials, equipment, processes, and procedures; shall be submitted in accordance with contract requirements. The sample size shall be as required by Table III.

4.3.1 First article inspection.-Prior to submission of the First Article to the Government, the contractor shall inspect components and assemblies for all contract, drawing, and specfication requirements. (See Table III). The contractor's inspection shall be subject to verification by the Government representative. Test equipment shall be the same as used in contract production.

4.3.1.1 First article test report.-The contractor shall prepare, in accordance with MIL-STD-831, six copies of DD Form 1222, appropriately completed by the Government Representative, and submitted with the sample. The DD Form 1222 shall be marked "First Article Test Report" and shall specify the applicable drawing numbers and revision dates, contract number, contractor's name and contract administrative office. This report shall contain, or have attached thereto, statements of findings covering mechanical properties on material covered by the drawings, and all examination and test results attesting conformance of the First Article to specification requirements.

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4.3.1.2 Examination and tests.-The First article sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility, specified in the contract. (See Table III).

4.3.1.3 First article sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval. No re-inspection shall be permitted on the original first article sample.

TABLE III

First Article Inspection

Examination or Test	Requirements	Method Para.	Number of Samples
Primer sensitivity	Spec. para. 3.4	Spec. para. 4.4.3.5	25
Sealant application	Dwg. D8594675	Spec. para. 4.4.3.6	1/
Protective finish	Spec. para. 3.2	Spec. para. 4.4.3.1	2/
Vibration	Spec. para. 3.3	Spec. para. 4.4.3.4	3/ 12
Functional	Spec. para. 3.7	Spec. para. 4.4.3.7	30 Delay Elements 30 Cartridges
Dimensional	Dimensions on all drawings		10 sets of unassembled components
Hardness	Dwg.	Spec. para. 4.4.3.2	2/
Metal defects	Spec. para. 3.8.1	Spec. para. 4.4.2	Same as for dimensional
Burrs	Spec. para. 3.8.2	Spec. para. 4.4.2	Same as for dimensional
Foreign matter	Spec. para. 3.8.3	Spec. para. 4.4.2	Same as for dimensional

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TABLE III (Cont'd)

1/ Run examination concurrently with primer sensitivity test using the same sample.

2/ Sample for dimensional examination shall be used for hardness tests and protective finish.

3/ To be taken from the cartridge ballistic sample.

4.4 Quality conformance inspection.

4.4.1 Lot.

4.4.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.4.1.2 Size of lot.-An ammumition lot shall consist of an identifiable group of metal parts assemblies consisting of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Primers from not more than one lot.

d. Igniters from not more than one lot.

e. Delay elements from not more than one lot.

4.4.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class the criteria in Table V shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ANSI-B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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TABLE IV

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE V

	Total Allowable Defects Per Sample Size			
Sample Size for Major Defects - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects - - - - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.4.2.1 Body, Delay, Dwg. C8594081, covering a detail of Dwg. D11730536.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Diameter of flash hole	Gage

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4.4.2.1 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of body with pitch diameter of thread	Gage
104. Diameter of cavity, max.	Gage
105. Diameter of body, min.	Gage
106. Diameter of thread relief	Gage
107. Diameter of step at thread relief	Gage
108. Depth of cavity, max.	Gage
109. Length from cavity face to body diameter	Gage
110. Thread damaged or not full	Visual
111. Protective finish inadequate or defective	Visual
112. Burr	Visual
113. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of wrench holes	Gage
203. Diameter of thread pilot	Gage
204. Diameter of cavity, min.	Gage
205. Diameter of body, max.	Gage
206. Depth of wrench holes	Gage
207. Depth of cavity, min.	Gage
208. Total length	Gage
209. Length of step at thread relief	Gage
210. Length of thread pilot	Gage
211. Length of thread relief	Gage
212. Location of wrench holes	Gage
213. Radius or chamfer missing or incorrect	Visual
214. Surface finish improper	Visual

4.4.2.2 Gasket, Aluminum, Dwg. B8594082, covering a detail of Dwg. D11730536.

CRITICAL

None defined

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4.4.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Diameter, outside, min.	Gage
102. Diameter, inside, max.	Gage
103. Metal defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, outside, max.	Gage
202. Diameter, inside, min.	Gage
203. Thickness	Gage
204. Surface finish improper	Visual

4.4.2.3 Cup, Retaining, Dwg. B8597232 covering a detail of Dwg. D11730536.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of counterbore with small hole	Gage
102. Concentricity of body with small hole	Gage
103. Diameter of counterbore, max.	Gage
104. Diameter of small hole, max.	Gage
105. Diameter of body, min.	Gage
106. Depth of counterbore	Gage
107. Surface finish of small hole and body improper	Visual
108. Burr or sharp corner	Visual
109. Foreign matter	Visual
MINOR	
201. Diameter of counterbore, min.	Gage
202. Diameter of small hole, min.	Gage
203. Diameter of body, max.	Gage
204. Total length	Gage
205. Surface finish improper	Visual

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4.4.2.4 Plug, Firing, Dwg. B8597234, covering a detail of Dwg. D11730536.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Diameter of flange, min.	Gage
103. Diameter of body, min.	Gage
104. Length of striker	Gage
105. Radius of striker	Gage
106. Surface finish of body improper	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter of flange, max.	Gage
202. Diameter of body, max.	Gage
203. Total length	Gage
204. Length of body	Gage
205. Radius missing or incorrect	Visual
206. Surface finish improper	Visual

4.4.2.5 Retainer, Primer, Drawing C11730537, covering a detail of Drawing D11730536.

CRITICAL	
1. Depth of primer pocket	Gage
2. Depth of cup pocket	Gage
3. Metal defective	Visual
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of flange with primer pocket	Gage
104. Concentricity of flash hole with primer pocket	Gage
105. Concentricity of cup pocket with primer pocket	Gage
106. Concentricity of body with primer pocket	Gage

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4.4.2.5 (Cont'd)

Defect	Method of Inspection
107. Concentricity of pitch diameter of thread with body	Gage
108. Symmetry of wrench flats with pitch diameter of threads	Gage
109. Diameter of body, min.	Gage
110. Diameter of primer pocket, min.	Gage
111. Diameter of cup pocket, min.	Gage
112. Length of effective thread, min.	Gage
113. Location of flange	Gage
114. Location of datum diameter	Gage
115. Thread damaged or not full	Visual
116. Protective finish inadequate or defective	Visual
117. Surface finish of primer pocket improper	Visual
118. Burr or sharp corner	Visual
119. Foreign matter	Visual

MINOR

201. Pitch diameter of thread, min.	Gage
202. Diameter of body, max.	Gage
203. Diameter of primer pocket, max.	Gage
204. Diameter of cup pocket, max.	Gage
205. Diameter of flange	Gage
206. Depth to bottom of threaded cavity	Gage
207. Length of wrench flats	Gage
208. Dimension across wrench flats	Gage
209. Total length	Gage
210. Location of wrench flats	Gage
211. Radius or chamfer missing or incorrect	Visual
212. Surface finish improper	Visual

4.4.2.6 Primer, Percussion, M42, Drawing B8799925, covering a detail of Drawing D11730536.

CRITICAL

1. Improper assembly	Visual
2. Any part missing	Visual
3. Primer pellet broken, cracked or dislodged	Visual
4. Metal defective	Visual
5. Primer foil discolored	Visual
6. Mixed primers	Visual

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4.4.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.4.2.7 Delay Element, Assembly Drawing 11730536 covering a detail of Drawing D8590475.

CRITICAL	
1. Improper assembly	X-Ray 1/ & Visual
2. Any part missing	X-Ray 1/ & Visual
3. Primer not seated to full depth of primer pocket	X-Ray 1/ & Visual
4. Delay column broken, cracked or dislodged	X-Ray 1/ & Visual
5. Flash hole missing or obstructed	Visual
6. Delay charge not visible through flash hole	Visual
7. Retainer cup not flush or below surface "S"	Visual
8. Firing plug not flush or below retainer cup	Visual
9. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Junction of delay body, gasket and primer retainer, not properly seated	Visual
103. Primer not properly crimped and sealed	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Total length, max.	Gage

1/ The assembled delay element shall be x-rayed with the long axis of the delay element perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

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4.4.2.8 Case, Cartridge, Dwg. C8595393 covering a detail of Dwg. D8594075.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of head, min.	Gage
104. Thickness of wall, min.	Gage
105. Thickness of base, min.	Gage
106. Protective finish inadequate or defective	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter of head, max.	Gage
202. Diameter of body	Gage
203. Total length	Gage
204. Thickness of wall, max.	Gage
205. Thickness of base, max.	Gage
206. Location of datum diameter	Gage
207. Radius missing or incorrect	Visual
208. Surface finish improper	Visual

4.4.2.9 Cartridge, Delay, M46 Assembly Drawing D8594075.

CRITICAL	
1. Any part missing	X-Ray 1/ & Visual
2. Improper assembly	X-Ray 1/ & Visual
3. Primer pellet broken, cracked, or dislodged	X-Ray 1/ & Visual
4. Delay column broken, cracked, or dislodged	X-Ray 1/ & Visual
5. Mixed delay elements or propellant	Visual
6. Metal defective	Visual
7. Crimp missing or incorrect	Visual
8. "O" ring split, cracked or cut	Visual
9. Marking missing, incorrect or illegible	Visual

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4.4.2.9 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Concentricity of cartridge head with body	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Length to datum diameter, max.	Gage
105. Cartridge profile failure	Gage
106. Firing plug or retaining cup above flush with head face	Visual
107. Protective finish inadequate or defective	Visual
108. Foreign matter	Visual
MINOR	
201. Total length	Gage
202. Fold, wrinkle, dent, buckle, bulge or scratch	Visual

1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.4.2.10 Carton, Packing, Ammunition for Cartridge, Delay M46, Drawing C8845190.

CRITICAL

None defined

MAJOR

None defined

MINOR

201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

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4.4.2.11 Carton, Alternative, Packing Ammunition, Drawing C9227731.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Cut or damaged through impregnated carton	Visual
203. Contents of box exposed	Visual
204. Carton damaged	Visual
205. Marking missing, incorrect or illegible	Visual
206. Number of cartons incorrect	Visual

4.4.2.12 Box, Packing, Ammunition Drawing C8845191.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of carton incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

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4.4.3 Test methods and procedures.

4.4.3.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in the determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with Table IV or V and MIL-C-60539. Test procedures for protective finish shall be in accordance with MIL-C-60539. The anodized parts shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.3.2 Hardness.-Ten units shall be selected for this test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot. The hardness test shall be performed on all heat-treated components prior to the application of protective finish. A hardness testing machine shall be used to determine compliance with the drawing requirements.

4.4.3.3 Airtightness.-Each assembled cartridge shall be subjected to the airtightness test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge. The method for determining airtightness shall be by use of the equipment listed on IEL 8594075.

4.4.3.4 Vibration.-Twelve of the 30 assembled x-rayed cartridges of the first article sample properly identified shall be subjected to vibration as in 3.3. After the vibration test, the twelve cartridges shall be x-rayed as directed in Note 1/ of 4.4.2.9. Evidence of any disintegration of primer, propellant, igniter, or delay charge or derangement of any component shall be cause for rejection of the lot. The twelve vibrated cartridges complying with 3.3, shall be grouped and temperature conditioned as indicated in Table VI and tested in accordance with 3.7.4. When a change occurs in the lot of primers, propellant, igniter or delay elements after the initiation of production, 12 assembled cartridges from the first production lot shall be tested in accordance with this paragraph. The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

TABLE VI

Temperature	Not Vibrated	Vibrated
-65°F to -70°F	6	4
65°F to 75°F	6	4
160°F to 165°F	6	4

4.4.3.5 Primer sensitivity.-A sample of 25 assembled cartridges from each production lot shall be tested for sensitivity. Failure of any primer to comply with the requirements of 3.4 shall be cause for rejection of the lot. Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge case around its periphery 1 inch maximum above the base. Drop test 25 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot at 13 inches, using a steel ball 15/16 inches in diameter, weighing 1.94 ounces plus or minus 0.02 ounces in a primer sensitivity machine with a cartridge holder, approved by the procuring activity.

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4.4.3.6 Sealant application.-Five cartridges from each production lot shall be utilized for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot. Sealant application examination to be run concurrently with primer sensitivity test; utilizing the same cartridges. The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of 360° of the joining surfaces at the junction of the "O" ring and case sidewall and that the propellant cavity is free of sealant.

4.4.3.7 Functional.-The following tests shall be performed by the Government:

4.4.3.7.1 Ballistics.-The 30 test delay elements, prior to cartridge assembly, shall be temperature conditioned as indicated in Table VII and 4.4.3.7.1.3. Failure of any delay element to comply with the requirements of 3.7.2, (Table I), shall be cause for lot rejection.

TABLE VII

Delay Element Assembly

Temperature	Sample Size
-65°F to -70°F	10
65°F to 75°F	10
160°F to 165°F	10

4.4.3.7.1.1 A sample of 30 assembled cartridges from each production lot shall be ballistically tested after temperature conditioning in accordance with Table VIII and 4.4.3.7.1.4. Failure of any cartridge to comply with the requirements of 3.7.4, (Table II), shall be cause for rejection of the lot.

TABLE VIII

Cartridge Assembly

Temperature Degree (°) Fahrenheit (°)	Sample
-65° to -70°	10
65° to 75°	10
160° to 165°	10

4.4.3.7.1.2 Ballistics.-The sample of delay elements and cartridges for ballistics tests shall be grouped and temperature conditioned in acccordance with 4.4.3.7.1.3, 4.4.3.7.1.4, Table VII and Table VIII, and rigidly supported in a test fixture. The samples shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.3.7.1.3 and 4.4.3.7.1.4.

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4.4.3.7.1.3 Delay element temperature conditioning.-A standard inert delay element containing a thermocoupler shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one (1) hour longer than the time necessary to bring the element contained in the inert delay element to the required temperature. The vent hole of the delay element samples shall be sealed against the environmental atmosphere during the conditioning period.

4.4.3.7.1.4 Cartridge temperature conditioning.-Cartridges shall be loaded into acceptable M4 or M6A1 Initiators with end caps to cover the gas ports during the conditioning period. The temperature of the propellant within the cartridges shall be considered as the conditioning temperature and will be determined by means of a thermocoupler attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the initiator undergoing tests. The conditioning time shall be 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

4.4.4 Inspection equipment.-The examinations and tests shall be performed using equipment listed on IEL 8594075.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring activity: Level "A" - Drawings C8845190 and C8845191. Level "B" Drawings C8845190 and C9227731.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Advisory delay element output material (Formulation).-Approximately 23.7 percent Boron, 70.7 percent Potassium Nitrate and 5.6 percent binder. Advisory individual delay element output material weight: approximately 1.5 grains.

6.3 Advisory delay element charge composition (Formulation).-Approximately 96.2 percent Barium Chromate and 3.8 percent Boron. Advisory individual delay element charge weight: approximately 21 grains.

6.4 Delay element assembly.-Delay time is defined as the time lapse between the impact of the firing pin on the primer and evidence of gas flow from the venthole of the delay element.

6.5 Cartridge.-Delay time is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

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6.6 Loaded Assembly Hazard Information:

Explosive Hazard Classification: Class 3
Storage Compatibility: Group B
Department of Transportation (DOT) Class C
DOT Marking - Small Arms Ammunition

6.7 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1377-A566

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-46272A(MU) Cartridge, Delay, M46 Assembly

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 65 WHICH MAY BE USED

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